

Fundamental Partial Differential Equations

Extention to 2D

Abstract

This report presents the extension of one-dimensional partial differential equations to two spatial dimensions. Building on prior work with convection, diffusion, and Burgers' equation, we introduce multi-dimensional fields, vector-valued solutions, and coupled systems. The transition to two dimensions represents a critical step toward realistic fluid flow modelling and the Navier–Stokes equations.

1. Introduction

In earlier studies, all equations were defined over a single spatial variable:

$$u = u(x, t)$$

However, most physical systems evolve over multiple spatial dimensions. In fluid mechanics, the velocity field depends on both horizontal and vertical coordinates:

$$u = u(x, y, t), \quad v = v(x, y, t)$$

The extension to two dimensions fundamentally alters:

- the structure of the equations
- the complexity of the numerical method
- the interpretation of physical behaviour

2. Two-Dimensional Fields

In two dimensions, the solution becomes a field defined over a grid:

$$(x_i, y_j), \quad i = 0, \dots, N_x, \quad j = 0, \dots, N_y$$

Numerically:

- $u(x, y, t)$ becomes a **2D array**
- spatial derivatives must be computed in both directions

3. Linear Convection in Two Dimensions

Governing equation

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} + c \frac{\partial u}{\partial y} = 0$$

Discretisation

Using forward time and backward space:

$$u_{i,j}^{n+1} = u_{i,j}^n - c \frac{\Delta t}{\Delta x} (u_{i,j}^n - u_{i-1,j}^n) - c \frac{\Delta t}{\Delta y} (u_{i,j}^n - u_{i,j-1}^n)$$

Interpretation

- The solution propagates diagonally through the domain
- The shape is preserved (as in 1D)

4. Nonlinear Convection in Two Dimensions

Governing equations

$$\begin{aligned} \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} &= 0 \\ \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} &= 0 \end{aligned}$$

Key features

- Two coupled equations for (u, v)
- Velocity field advects itself
- Strong nonlinearity

Interpretation

- Flow direction depends on local velocity
- Distortion and steepening occur in multiple directions

5. Diffusion in Two Dimensions

Governing equation

$$\frac{\partial u}{\partial t} = \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

Discretisation

$$u_{i,j}^{n+1} = u_{i,j}^n + \nu \frac{\Delta t}{\Delta x^2} (u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n) + \nu \frac{\Delta t}{\Delta y^2} (u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n)$$

Interpretation

- Diffusion spreads in all spatial directions
- Peaks flatten radially (not just along one axis)

6. Two-Dimensional Burgers' Equation

Governing equations

$$\begin{aligned}\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} &= \nu \nabla^2 u \\ \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} &= \nu \nabla^2 v\end{aligned}$$

Interpretation

- Combines:
 - nonlinear transport (steepening)
 - diffusion (smoothing)
- Acts as a simplified model of viscous flow

7. Mathematical Formulation and Implementation Roadmap for Two-Dimensional PDE Solvers

7.1 Objective

The purpose of this section is to describe, in mathematical terms, how the one-dimensional model equations introduced earlier are extended to two spatial dimensions.

The main transition is from a scalar field in one space dimension,

$$u = u(x, t),$$

to fields defined over a two-dimensional spatial domain,

$$u = u(x, y, t).$$

For fluid-flow problems, the unknown is typically a velocity vector field,

$$\mathbf{u}(x, y, t) = \begin{bmatrix} u(x, y, t) \\ v(x, y, t) \end{bmatrix},$$

where u is the velocity component in the x -direction and v is the velocity component in the y -direction.

This extension is the essential intermediate step between one-dimensional convection--diffusion problems and the full incompressible Navier--Stokes equations.

7.2 Computational Domain

Let the two-dimensional spatial domain be

$$\Omega = [0, L_x] \times [0, L_y].$$

The domain is discretised using a uniform Cartesian grid:

$$x_i = i\Delta x, \quad y_j = j\Delta y,$$

where

$$i = 0, 1, \dots, N_x - 1, \quad j = 0, 1, \dots, N_y - 1.$$

The spatial step sizes are

$$\Delta x = \frac{L_x}{N_x - 1}, \quad \Delta y = \frac{L_y}{N_y - 1}.$$

Time is discretised as

$$t^n = n\Delta t,$$

where Δt is the time step and n is the time-level index.

The discrete approximation to a scalar field is written as

$$u(x_i, y_j, t^n) \approx u_{i,j}^n.$$

7.3 Two-Dimensional Linear Convection

The two-dimensional linear convection equation is

$$\frac{\partial u}{\partial t} + c_x \frac{\partial u}{\partial x} + c_y \frac{\partial u}{\partial y} = 0,$$

where c_x and c_y are constant transport velocities.

Using a forward difference in time and backward differences in space gives

$$\frac{u_{i,j}^{n+1} - u_{i,j}^n}{\Delta t} + c_x \frac{u_{i,j}^n - u_{i-1,j}^n}{\Delta x} + c_y \frac{u_{i,j}^n - u_{i,j-1}^n}{\Delta y} = 0.$$

Solving for the next time level gives

$$u_{i,j}^{n+1} = u_{i,j}^n - c_x \frac{\Delta t}{\Delta x} (u_{i,j}^n - u_{i-1,j}^n) - c_y \frac{\Delta t}{\Delta y} (u_{i,j}^n - u_{i,j-1}^n).$$

This equation describes the translation of an initial profile through the two-dimensional domain.

7.4 Two-Dimensional Nonlinear Convection

For a two-dimensional velocity field, nonlinear convection is described by the coupled system

$$\begin{aligned} \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} &= 0, \\ \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} &= 0. \end{aligned}$$

Equivalently, in vector notation,

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = \mathbf{0}.$$

The discrete update for the u -component is

$$u_{i,j}^{n+1} = u_{i,j}^n - u_{i,j}^n \frac{\Delta t}{\Delta x} (u_{i,j}^n - u_{i-1,j}^n) - v_{i,j}^n \frac{\Delta t}{\Delta y} (u_{i,j}^n - u_{i,j-1}^n).$$

The discrete update for the v -component is

$$v_{i,j}^{n+1} = v_{i,j}^n - u_{i,j}^n \frac{\Delta t}{\Delta x} (v_{i,j}^n - v_{i-1,j}^n) - v_{i,j}^n \frac{\Delta t}{\Delta y} (v_{i,j}^n - v_{i,j-1}^n).$$

This system is important because the same nonlinear convective structure appears in the Navier--Stokes momentum equations.

7.5 Two-Dimensional Diffusion

The two-dimensional diffusion equation is

$$\frac{\partial u}{\partial t} = \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right),$$

or, using the Laplacian operator,

$$\frac{\partial u}{\partial t} = \nu \nabla^2 u.$$

Using forward Euler in time and central differences in space gives

$$\frac{u_{i,j}^{n+1} - u_{i,j}^n}{\Delta t} = \nu \left[\frac{u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{\Delta x^2} + \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{\Delta y^2} \right].$$

Therefore,

$$u_{i,j}^{n+1} = u_{i,j}^n + \nu \frac{\Delta t}{\Delta x^2} (u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n) + \nu \frac{\Delta t}{\Delta y^2} (u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n).$$

This equation models smoothing in both spatial directions.

7.6 Two-Dimensional Burgers' Equations

The two-dimensional Burgers' equations combine nonlinear convection and diffusion:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right),$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right).$$

In vector notation,

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = \nu \nabla^2 \mathbf{u}.$$

The discrete update for the u -component is

$$\begin{aligned} u_{i,j}^{n+1} = & u_{i,j}^n - u_{i,j}^n \frac{\Delta t}{\Delta x} (u_{i,j}^n - u_{i-1,j}^n) - v_{i,j}^n \frac{\Delta t}{\Delta y} (u_{i,j}^n - u_{i,j-1}^n) \\ & + \nu \frac{\Delta t}{\Delta x^2} (u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n) + \nu \frac{\Delta t}{\Delta y^2} (u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n). \end{aligned}$$

Similarly, the update for the v -component is

$$\begin{aligned} v_{i,j}^{n+1} = & v_{i,j}^n - u_{i,j}^n \frac{\Delta t}{\Delta x} (v_{i,j}^n - v_{i-1,j}^n) - v_{i,j}^n \frac{\Delta t}{\Delta y} (v_{i,j}^n - v_{i,j-1}^n) \\ & + \nu \frac{\Delta t}{\Delta x^2} (v_{i+1,j}^n - 2v_{i,j}^n + v_{i-1,j}^n) + \nu \frac{\Delta t}{\Delta y^2} (v_{i,j+1}^n - 2v_{i,j}^n + v_{i,j-1}^n). \end{aligned}$$

Burgers' equations are a useful intermediate model because they contain the nonlinear convection and viscous diffusion terms of the Navier–Stokes equations, but do not yet include pressure.

7.7 Boundary Conditions

A numerical solution requires boundary conditions on the boundary of the domain,

$$\partial \Omega.$$

A Dirichlet boundary condition prescribes the value of the solution:

$$u = g(x, y, t) \quad \text{on } \partial\Omega.$$

A Neumann boundary condition prescribes the normal derivative:

$$\frac{\partial u}{\partial n} = h(x, y, t) \quad \text{on } \partial\Omega.$$

For vector-valued problems, boundary conditions must be specified for both components:

$$u = g_1(x, y, t), \quad v = g_2(x, y, t) \quad \text{on } \partial\Omega.$$

The choice of boundary conditions is problem-dependent and has a significant influence on the resulting solution.

7.8 Stability Conditions

Explicit finite-difference schemes require restrictions on the time step.

For two-dimensional convection, the CFL condition may be written as

$$\max |u| \frac{\Delta t}{\Delta x} + \max |v| \frac{\Delta t}{\Delta y} \leq 1.$$

For linear convection with constant speeds c_x and c_y , this becomes

$$|c_x| \frac{\Delta t}{\Delta x} + |c_y| \frac{\Delta t}{\Delta y} \leq 1.$$

For diffusion, a common explicit stability condition is

$$\nu \Delta t \left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} \right) \leq \frac{1}{2}.$$

For Burgers' equations, the time step must satisfy both convection and diffusion constraints.

7.9 Connection to the Navier--Stokes Equations

The incompressible Navier--Stokes equations in two dimensions are

$$\begin{aligned} \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \\ \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} &= -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right), \end{aligned}$$

together with the incompressibility condition

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

Comparing Burgers' equations with Navier--Stokes shows that Navier--Stokes adds two essential features:

1. a pressure field $p(x, y, t)$,
2. the incompressibility constraint.

The pressure is not merely an additional unknown. It acts as a constraint-enforcing variable that ensures the velocity field remains divergence-free.